



MSC IN MATHEMATICAL ENGINEERING A.Y. 2025/2026
PROJECT REPORT OF BAYESIAN STATISTICS (052499) – PROF. A. GUGLIELMI
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Clustering clusters with exogenous information

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19th February 2026

Abstract

In many scenarios, while performing statistical analysis, observations can be divided in homogeneous groups. Groups obey some similarity dictated by a modeling assumption, and heterogeneity is encouraged between distinct groups. However, groups can be similar as well on a latent level. In this project we explore a model to perform informed clustering of clusters. Ideally, once that observations are divided in clusters, those clusters are then grouped together on the base of some exogenous information through covariate measurements. The project is motivated by an astronomical study: we consider stars belonging to the same galaxy, for which we want to identify clusters related to distinct structural parts, namely arm and core areas of the galaxy. However, groups of stars belonging to different arms share similar dynamics in the galaxy, and should be clustered together on a higher level.

The source code of the entire project,
including this report and the presentations, is available at
 [alessio6666/Bayesian-Statistics](https://github.com/alessio6666/Bayesian-Statistics)

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1 Introduction

Clustering large astronomical datasets is challenging because stellar structures can be highly complex, and spatial coordinates alone may not be sufficient to capture all relevant patterns. This study considers the Product Partition Model (PPM), a Bayesian approach that defines clusters through cohesion functions under an exchangeability assumption. To address the limitations of relying only on spatial information, the Product Partition Model with Covariates (PPMX) is also examined. This extension incorporates a similarity function into the prior distribution of the partition, allowing external covariates to directly influence cluster formation. The analysis is motivated by data from the LISA mission and focuses on a subset of stars within a single galaxy, rather than the full dataset of 32 million observations. In this setting, stars located in different spiral arms may exhibit similar dynamical properties, making spatial information alone potentially insufficient to identify meaningful structures. For this reason, physically relevant covariates specifically frequency and frequency derivatives are included to assess how their incorporation affects clustering results compared to a purely spatial PPM, without assuming in advance that additional information necessarily improves performance. The results indicate that including covariates leads to more parsimonious and stable partitions, reduces posterior uncertainty, and enhances physical interpretability.

2 Literature Review and Methodological Framework

The identification of latent structures in high dimensional datasets has evolved from rigid parametric clustering toward flexible Bayesian nonparametric (BNP) frameworks. The foundation of our research lies in the Product Partition Model (PPM) introduced by (Barry and Hartigan, 1992), which we describe in detail below.

2.1 Product Partition Model (PPM)

The PPM is a Bayesian nonparametric framework that places a prior directly on the space of partitions of the observational units. The model represents the data generating mechanism through a random clustering structure, where observations assigned to the same cluster share a common parameter.

2.1.1 Data and Notation

Let $\mathbf{y}_1, \dots, \mathbf{y}_n$ denote the observed data. Each observation is associated with a latent cluster label $c_i \in \{1, 2, \dots\}$, and the resulting partition of the index set $\{1, \dots, n\}$ is denoted by

$$\rho_n = \{S_1, \dots, S_k\},$$

where S_j are disjoint subsets whose union is $\{1, \dots, n\}$. We denote by $n_j = |S_j|$ the size of cluster S_j , by θ_j the parameter associated with cluster S_j , and by $p(\rho_n)$ the prior distribution over partitions.

2.1.2 Model Assumptions

A1: Likelihood given the clustering. For each cluster S_j , conditional on the cluster parameter θ_j , the observations within each cluster are conditionally independent:

$$y_i \mid c_i = j, \theta_j \stackrel{\text{ind}}{\sim} f(y_i \mid \theta_j), \quad i \in S_j.$$

A2: Prior distribution for the cluster parameters. Cluster specific parameters are assumed independent and identically distributed from a base measure P_0 :

$$\theta_j \stackrel{\text{iid}}{\sim} P_0.$$

A3: Prior on the partition with a product structure. The PPM prior on partitions is defined as

$$p(\rho_n) \propto \prod_{j=1}^k c(S_j),$$

where $c(S_j)$ denotes the cohesion function associated with subset S_j . Under A1–A3, the full hierarchical model of the PPM is

$$\begin{aligned}\rho_n &\sim p(\rho_n), \\ \theta_j \mid \rho_n &\stackrel{\text{iid}}{\sim} P_0, \quad j = 1, \dots, k, \\ y_i \mid c_i = j, \theta_j &\stackrel{\text{iid}}{\sim} f(y_i \mid \theta_j), \quad i \in S_j.\end{aligned}$$

While the PPM provides a robust mechanism for partitioning observations via cluster-specific cohesions, it inherently assumes exchangeability. In the context of spatially indexed astronomical data, such as the coordinates provided by the LISA mission, this assumption can be overly restrictive, as it does not incorporate covariate-dependent structure within the partition prior.

2.1.3 Pitman–Yor Process and Induced Cohesion

A widely adopted choice for the cohesion function $c(\cdot)$ is the one induced by a Dirichlet Process (DP) or, more generally, by a Pitman–Yor (PY) process. Although the DP is the most widely adopted prior in BNP models, De Blasi et al. (2015) show that its cluster growth rate may be restrictive in large, heterogeneous systems. The PY process generalizes the DP by introducing a discount parameter that allows for power-law growth in the number of clusters, a property particularly relevant in modeling stellar populations of the Milky Way, which exhibit heavy-tailed structural behavior.

In the PY framework, a random probability measure is defined as

$$G \sim \text{PY}(\alpha, d, G_0),$$

where $\alpha > -d$ is the concentration parameter, $0 \leq d < 1$ is the discount parameter, and G_0 is the base distribution. When $d = 0$, the PY process reduces to the Dirichlet Process. The induced predictive probability for the cluster allocation follows

$$P(c_i = j \mid c_{-i}) = \begin{cases} \frac{n_{-i,j} - d}{\alpha + n - 1}, & \text{existing cluster,} \\ \frac{\alpha + dk}{\alpha + n - 1}, & \text{new cluster,} \end{cases}$$

where $n_{-i,j}$ is the size of cluster j excluding observation i , and k is the number of non-empty clusters in c_{-i} .

2.1.4 Full Conditional for Cluster Assignments

Posterior inference is performed via Gibbs sampling by updating each allocation variable c_i conditionally on the remaining labels and parameters. In the normal–normal conjugate setting, the full conditional takes the following form.

Allocated cluster. For an existing cluster j ,

$$P(c_i = j \mid \cdot) \propto (n_{-i,j} - d) \mathcal{N}(y_i \mid m_j, \sigma^2 + \tau_j^2),$$

where m_j and τ_j^2 denote the posterior mean and variance of θ_j computed from the observations currently assigned to cluster j (excluding i).

New cluster. For assignment to a new cluster,

$$P(c_i = \text{new} \mid \cdot) \propto (\alpha + dk) \mathcal{N}(y_i \mid m_0, \sigma^2 + \tau_0^2),$$

where m_0 and τ_0^2 correspond to the prior predictive quantities implied by P_0 . Assuming

$$y_i \mid \theta_j \sim \mathcal{N}(\theta_j, \sigma^2), \quad \theta_j \sim \mathcal{N}(m_0, \tau_0^2),$$

the posterior distribution of θ_j given the observations in cluster j (excluding i) is Gaussian with parameters

$$\tau_j^2 = \left(\frac{n_{-i,j}}{\sigma^2} + \frac{1}{\tau_0^2} \right)^{-1},$$

$$m_j = \tau_j^2 \left(\frac{\sum_{l \in S_j \setminus \{i\}} y_l}{\sigma^2} + \frac{m_0}{\tau_0^2} \right).$$

2.2 From PPM to PPMX

The PPM framework models clustering entirely through the response $\mathbf{y}_1, \dots, \mathbf{y}_n$. The partition prior $p(\rho_n) \propto \prod_{j=1}^k c(S_j)$ depends only on the cluster configuration, making the induced partition structure exchangeable with respect to the observations. In many practical applications, however, each observation is accompanied by covariates that carry relevant structural information. In astronomical contexts, spatial proximity alone is often insufficient to distinguish between overlapping physical dynamics of galaxy arms and core regions. While in the classical PPM such covariates may enter the likelihood, they do not influence the prior distribution over partitions. To address these limitations, Müller, Quintana and Rosner (2011) developed the Product Partition Model with Covariates (PPMX), which extends the PPM by incorporating a similarity function $g(\mathbf{x}_{S_j})$ directly within the partition prior. This formulation allows exogenous physical information such as frequency and frequency derivatives in our dataset to influence the clustering structure, representing a structural enhancement rather than merely a computational modification.

2.3 Product Partition Model with Covariates (PPMX)

2.3.1 Model Specification

In the PPMX, we consider observations $\mathbf{y}_1, \dots, \mathbf{y}_n$ with associated covariates $\mathbf{x}_1, \dots, \mathbf{x}_n$. Let $\rho_n = \{S_1, \dots, S_k\}$ be a partition of $\{1, \dots, n\}$. The joint distribution is defined by three components:

1. Partition Prior. The prior on the partition is weighted by a similarity function $g(\mathbf{x}_{S_j})$ acting on the covariates within each cluster:

$$p(\rho_n \mid \mathbf{x}_{1:n}) \propto \prod_{j=1}^k c(S_j) g(\mathbf{x}_{S_j}),$$

where $c(S_j)$ is the cohesion function and $g(\mathbf{x}_{S_j})$ quantifies covariate-dependent similarity. In contrast to the PPM, the partition prior in the PPMX is no longer exchangeable in the observations.

2. Cluster Parameters. For each cluster j , parameters θ_j are drawn independently:

$$\theta_j \stackrel{\text{iid}}{\sim} P_0.$$

3. Likelihood. Conditional on the partition and cluster parameters:

$$y_i \mid c_i = j, \theta_j \stackrel{\text{iid}}{\sim} f(y_i \mid \theta_j), \quad i \in S_j.$$

The full hierarchical model is therefore

$$\begin{aligned} \rho_n \mid \mathbf{x}_{1:n} &\sim p(\rho_n \mid \mathbf{x}_{1:n}), \\ \theta_j \mid \rho_n &\stackrel{\text{iid}}{\sim} P_0, \quad j = 1, \dots, k, \\ y_i \mid c_i = j, \theta_j &\stackrel{\text{iid}}{\sim} f(y_i \mid \theta_j), \quad i \in S_j. \end{aligned}$$

2.3.2 Similarity Function

The function $g(\mathbf{x}_{S_j})$ measures covariate similarity among observations within cluster S_j and is typically defined as the marginal likelihood of an auxiliary model $q(\cdot)$:

$$g(\mathbf{x}_{S_j}) = \int \left[\prod_{i \in S_j} q(\mathbf{x}_i \mid \xi_j) \right] q(\xi_j) d\xi_j.$$

Common choices for $q(\cdot)$ include Normal-Inverse-Gamma for continuous covariates or Dirichlet-Categorical for categorical ones.

2.3.3 MCMC Update for Cluster Assignment

Cluster labels c_i are updated via a Gibbs sampling step. The full conditional for assigning observation i to cluster j is

$$P(c_i = j \mid \dots) \propto \underbrace{P(c_i = j \mid c_{-i})}_{\text{Partition Prior}} \cdot \underbrace{f(y_i \mid y_{S_j \setminus \{i\}})}_{\text{Predictive Density}} \cdot \underbrace{\frac{g(\mathbf{x}_{S_j \cup \{i\}})}{g(\mathbf{x}_{S_j})}}_{\text{Covariate Adjustment}},$$

where $f(y_i \mid y_{S_j \setminus \{i\}})$ is the posterior predictive density obtained by integrating out θ_j under P_0 . The three terms play distinct roles: the partition prior is governed by the Pitman–Yor process, the predictive density captures how well y_i fits the current cluster distribution, and the covariate adjustment modifies the assignment probability based on covariate similarity between $\mathbf{x}_i$ and the existing members of cluster j .

2.4 Scalability and Contribution

Despite these methodological developments, the scalable application of covariate-informed Pitman–Yor partition models to massive astronomical datasets (exceeding 30 million observations) remains largely unexplored. While the foundational sampling algorithms proposed by Neal (2000) provide a rigorous framework for inference in BNP mixture models, their direct implementation becomes computationally demanding at this scale. Our work contributes to this area by combining the covariate-driven flexibility of PPMX with the structural generality of the PY process, implemented through a strategic subsampling strategy to maintain computational feasibility. This synthesis enables a parsimonious and physically coherent clustering of stellar observations that integrates both spatial and physical characteristics.

3 Exploratory Data Analysis

Our analysis is based on data derived from the LISA (Laser Interferometer Space Antenna) mission, which is designed to detect and measure gravitational waves from astrophysical sources. The primary objective of our study is to apply Bayesian Clustering techniques to a high-dimensional astronomical dataset comprising approximately 32.7 million observations.

Each observation corresponds to a detected astrophysical source and is characterized by the following variables:

- **Cov1**: Frequency.
- **Cov2**: Frequency Derivative.
- **Latitude (b)**: Ecliptic latitude coordinates.
- **Longitude (l)**: Ecliptic longitude coordinates.

The frequency and its derivative describe the physical properties of the gravitational wave signal, while latitude and longitude provide the spatial position of the source.

3.1 Global Statistics

After a preliminary summary of the dataset we can observe that it contains 32,713,638 observations with no missing data points. We can also notice significant variance across the frequency parameters, As shown in Table 1.

Statistic	Cov1 (Frequency)	Cov2 (Freq. Deriv.)	Latitude (b)	Longitude (l)
Mean	2.59×10^{-4}	6.19×10^{-17}	0.0065	-1.3682
Variance	5.85×10^{-8}	5.93×10^{-32}	0.1743	0.5450
Minimum	9.26×10^{-5}	7.97×10^{-23}	-1.5658	-3.1416
Maximum	3.15×10^{-2}	6.76×10^{-13}	1.5643	3.1416
Total Observations	32,713,638			
Missing Values	0			

Table 1: Global descriptive statistics of the LISA dataset.

The frequency variable spans several orders of magnitude, ranging from approximately 10^{-5} to 10^{-2} . The frequency derivative is more concentrated around extremely small values, with magnitudes between 10^{-23} and 10^{-13} . These large scale differences suggest strong heterogeneity and can lead to transformation strategies.

Latitude and longitude exhibit ranges consistent with spherical coordinates: latitude is approximately bounded between $-\pi/2$ and $\pi/2$, while longitude spans nearly the full $[-\pi, \pi]$ interval.

3.2 Spatial Visualization

The spatial distribution of the data was analyzed using Ecliptic coordinates. Visualizing the global density map (log-scaled) reveals a highly non-uniform structure.

The analysis of the spatial plots highlights:

- A clear curved pattern across the ecliptic plan.
- A high concentration of sources in specific, well-defined high-density zones.
- Significant heterogeneity in the distribution of observations.

The presence of structured high-density areas suggests that the data naturally form clusters in space.

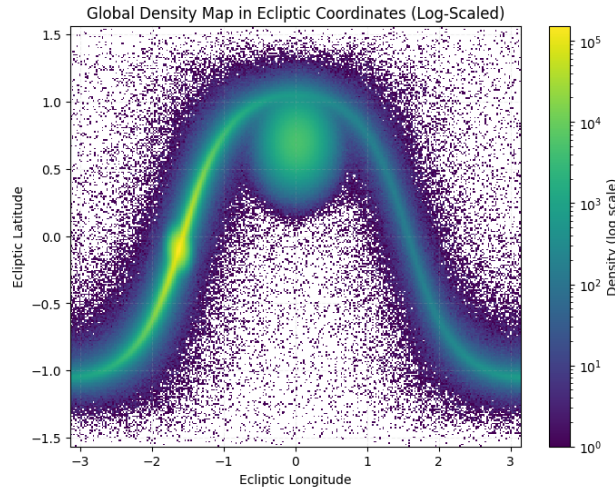


Figure 1: Global Density Map in Ecliptic Coordinates (Log-Scaled)

3.3 Univariate Distributions and Transformations

The univariate distributions of Longitude and Latitude show strong heterogeneity and are characterized by pronounced peaks. To better interpret these structures and handle the long-tailed nature of the data, we decided to apply a log-transformation.

3.3.1 Log-Transformation Impact

Applying the transformation $\log(\text{Coordinate} + \text{constant})$ yielded the following improvements:

- Tail Compression: The extreme tails of the original distributions were compressed, making the data more manageable for clustering algorithms.
- Peak Definition: Multiple peaks in both Latitude and Longitude became more defined and interpretable in the log-transformed space.

After transformation, the structure becomes substantially clearer. Peaks that were partially obscured in the original scale become more distinguishable in the log-domain. This suggests that the underlying generative structure of the data is more easily interpretable on a logarithmic scale.

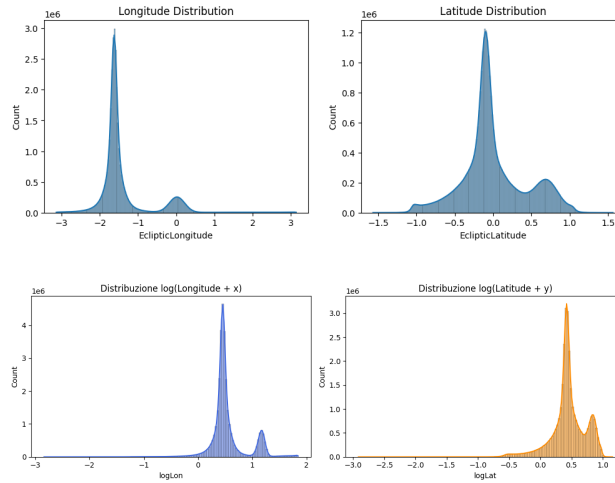


Figure 2: Longitude and Latitude distributions before and after log-transformation.

3.3.2 Bivariate Correlation in Transformed Space

After the transformation a much clearer curvilinear trend is revealed when comparing Longitude and Latitude. This high-density structure is a critical feature of the dataset that only emerges clearly following the log-transformation. The emergence of this pattern suggests that the clustering model must be capable of capturing non-linear relationships within the covariate space.

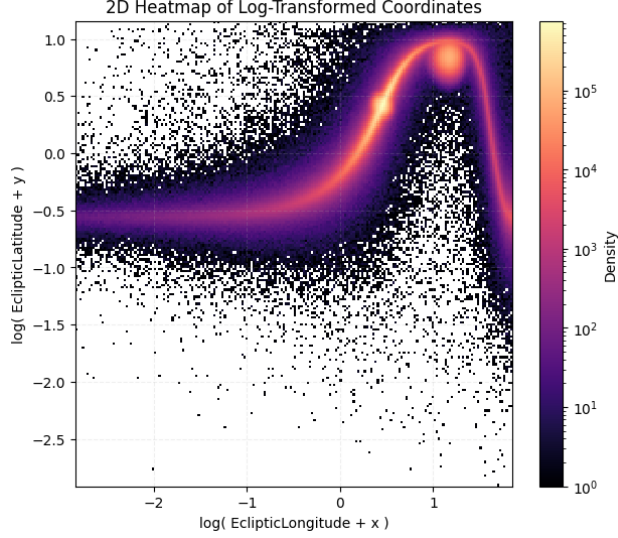


Figure 3: 2D Heatmap of Log-Transformed Coordinates showing the emerging curvilinear trend.

The exploratory analysis reveals the emergence of clearer geometric patterns after logarithmic transformation that justifies the adoption of advanced Bayesian nonparametric methods, such as the Product Partition Model (PPM), to effectively identify its latent cluster.

4 Methodology

4.1 Data Preprocessing and Subsampling

The full LISA dataset contains over 32 million observations, which renders direct application of the Gibbs sampler computationally prohibitive given its $O(N \cdot K)$ complexity per iteration. We therefore adopt a representative random subsampling strategy, drawing a subset of $n = 3,000$ observations with a fixed random seed to ensure reproducibility. The Pitman-Yor hyperparameters α and d are then tuned to maintain a clustering density consistent with the full population. The physical response variables (EclipticLatitude, EclipticLongitude) are standardized to zero mean and unit variance. For the covariates Cov1 (f_0 , gravitational wave frequency) and Cov2 ($\dot{f}$, its time derivative), which span several orders of magnitude, a logarithmic transformation is applied prior to standardization:

$$x_i^{(d)} \leftarrow \frac{\log x_i^{(d)} - \overline{\log x^{(d)}}}{\text{sd}(\log x^{(d)})}.$$

This renders the covariates approximately Gaussian, which is a necessary condition for the NIW similarity function used in the PPMX.

4.2 Model Specification and Implementation

We implement two models for comparison: a collapsed Pitman-Yor Gaussian mixture (PPM) and its covariate-extended version (PPMx). In both cases, the response is modelled as

$$y_i \mid c_i = j, \mu_j, \Sigma_j \sim \mathcal{N}_D(\mu_j, \Sigma_j),$$

with a Pitman-Yor process prior on the partition governed by parameters (α, d) .

The cluster parameters (μ_j, Σ_j) are assigned a Normal-Inverse-Wishart base measure G_0

$$(\mu_j, \Sigma_j) \sim \mathcal{N}\left(m_0, \frac{\Sigma_j}{\kappa_0}\right) \times \text{IW}(\nu_0, \Lambda_0).$$

with hyperparameters m_0 (prior mean), κ_0 (prior strength), ν_0 (degrees of freedom), and Λ_0 (scale matrix). Conjugacy allows (μ_j, Σ_j) to be analytically integrated out, yielding a *collapsed* Gibbs sampler that avoids sampling high-dimensional cluster parameters and substantially improves mixing. (Murphy, 2007)

Given cluster sufficient statistics $(n_j, \bar{y}_j, S_j)$, the NIW posterior parameters update as

$$\kappa_n = \kappa_0 + n_j, \quad \nu_n = \nu_0 + n_j, \quad m_n = \frac{\kappa_0 m_0 + n_j \bar{y}_j}{\kappa_n}, \quad \Lambda_n = \Lambda_0 + S_j + \frac{\kappa_0 n_j}{\kappa_n} (\bar{y}_j - m_0)(\bar{y}_j - m_0)^\top.$$

Due to conjugacy, marginalizing out (μ_j, Σ_j) yields a multivariate Student- t predictive distribution. The log-density implemented in `log_mvmt` is:

$$\log p(y_i) = \log \Gamma\left(\frac{\nu + D}{2}\right) - \log \Gamma\left(\frac{\nu}{2}\right) - \frac{D}{2} \log(\nu\pi) - \frac{1}{2} \log |\Sigma_n| - \frac{\nu + D}{2} \log\left(1 + \frac{D_M^2}{\nu}\right),$$

where the Mahalanobis distance $D_M^2 = (y_i - m_n)^\top \Sigma_n^{-1} (y_i - m_n)$ measures how far an observation lies from the cluster center while accounting for cluster covariance, ensuring that directions with high variance penalize distance less strongly. (Mahalanobis, 1936) The predictive scale matrix is

$$\Sigma_n = \frac{\kappa_n + 1}{\kappa_n(\nu_n - D + 1)} \Lambda_n,$$

and for a new cluster the density is evaluated using only the prior hyperparameters. All computations are performed in the log-domain to prevent numerical underflow, which occurs when multiplying the small probabilities of the prior weights and predictive densities. The final normalization of cluster assignment probabilities is handled via the Log-Sum-Exp trick to prevent numerical underflow. (Blanchard, D. J. Higham and N. J. Higham, 2021)

Each Gibbs iteration proceeds by: (1) removing observation i from its cluster, (2) updating sufficient statistics, (3) computing Pitman–Yor prior weights, (4) evaluating the Student- t predictive density, (5) multiplying and normalizing, and (6) sampling a new cluster label. This cycle is repeated for all observations across all iterations.

In the **PPMx** variant, the partition prior is further weighted by a covariate similarity term $g(\mathbf{x}_{S_j})$, implemented as the NIW marginal likelihood of the covariates within each cluster. The full conditional becomes

$$P(c_i = j \mid \text{rest}) \propto (n_{-i,j} - d) \cdot p(y_i \mid \mathbf{y}_{-i,j}) \cdot \left[\frac{g(\mathbf{x}_{S_j \cup \{i\}})}{g(\mathbf{x}_{S_j})} \right]^\eta,$$

where $\eta \geq 0$ scales covariate influence; $\eta = 0$ recovers the standard PPM.

4.3 Posterior Inference and Point Estimation

After running the Gibbs sampler, we compute the Posterior Similarity Matrix (PSM). The construction of the PSM is necessary to address the ‘label switching’ problem inherent in MCMC sampling for mixture models. Since the cluster indices (for example $c_i = 1$) are arbitrary and can change between iterations, the PSM provides a more label-invariant summary of the posterior by focusing on the pairwise co-clustering probability of observations, computed over the M post-burn-in samples.

$$\text{PSM}_{ij} = \frac{1}{M} \sum_{m=1}^M \mathbf{1}\{c_i^{(m)} = c_j^{(m)}\},$$

The final clustering is selected by minimizing the Binder loss relative to the PSM. The Binder loss function identifies the optimal point estimate of the partition by minimizing the posterior expected cost of misclustering. (Wade and Ghahramani, 2018)

$$L(\mathbf{c}) = \sum_{i < j} (\mathbf{1}\{c_i = c_j\} - \text{PSM}_{ij})^2,$$

This gives the partition that is closest to the posterior average clustering structure.

5 Results of Bayesian Clustering

5.1 MCMC Diagnostics: Number of Clusters and Partition Entropy

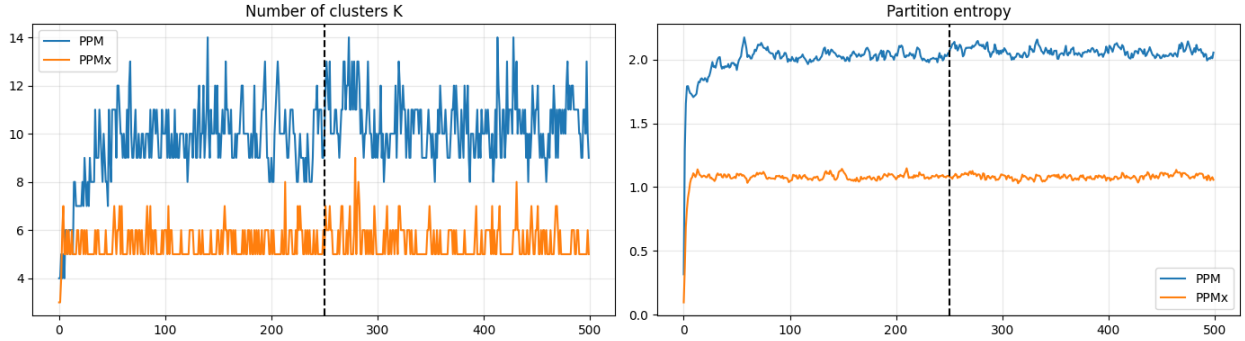


Figure 4: it reports the posterior evolution of two key quantities for both models: the number of clusters K (left panel) and the partition entropy (right panel). In both panels, the horizontal axis represents the MCMC iteration index ($t = 1, \dots, 500$), while the vertical dashed line marks the burn-in cutoff (first half of the chain).

5.1.0.1 Number of clusters K . The left panel displays the sampled number of clusters $K^{(t)}$ at each MCMC iteration. For the PPM model (blue curve), after burn-in the number of clusters fluctuates approximately between 8 and 13, with substantial variability across iterations. This behavior indicates that the posterior distribution is not concentrated on a single partition size, but instead spreads its probability across several plausible clustering configurations. In practical terms, the purely spatial model tends to generate a relatively fragmented clustering configuration and exhibits higher posterior uncertainty regarding the true partition. In contrast, the PPMX model (orange curve) stabilizes after burn-in between approximately 5 and 7 clusters, with markedly reduced variability. The oscillations are smaller and the chain appears more concentrated around a narrower range of K values. This suggests that incorporating covariate similarity leads to a more parsimonious and stable clustering structure.

5.1.0.2 Partition entropy. The right panel reports the partition entropy across iterations. Entropy here summarizes the overall complexity and dispersion of the clustering configuration: higher values indicate more fragmented and uncertain partitions, whereas lower values correspond to more coherent and concentrated structures. For the PPM model, entropy stabilizes around a higher level (approximately ~ 2.0), with visible fluctuations even after burn-in. This is consistent with the larger number of clusters and the higher variability observed in the left panel. For the PPMX model, entropy stabilizes around a substantially lower level (approximately ~ 1.1) and remains nearly flat after burn-in. This indicates stronger posterior agreement in cluster assignments and a more concentrated posterior distribution over partitions.

5.1.0.3 Comparison and interpretation. Taken together, the two panels clearly show that the transition from PPM to PPMX does not simply reduce the number of clusters, but also changes the global structure of posterior uncertainty. The PPM model, driven solely by spatial information, produces more clusters and higher entropy, reflecting greater fragmentation and variability across sampled partitions. The PPMX model, by incorporating covariate similarity, reduces both the posterior number of clusters and the partition entropy, leading to a more stable and coherent posterior clustering structure. Overall, the diagnostics indicate that incorporating covariate information decreases the posterior variance of the partition distribution and concentrates posterior mass around fewer, more structurally coherent clustering configurations.

5.2 Posterior Similarity Matrices

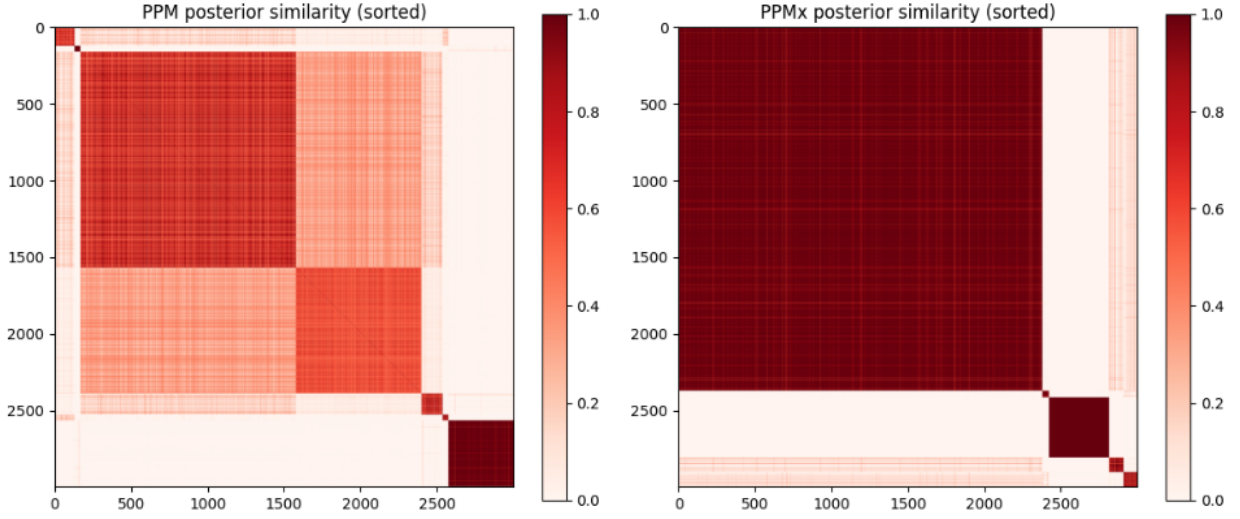


Figure 5: The posterior similarity matrices (PSM) provide a pairwise summary of clustering uncertainty under the PPM and PPMx models. For each pair of observations (i, j) , the matrix entry represents the posterior probability $P(z_i = z_j \mid \text{data})$, that is, the probability that the two observations belong to the same cluster. The matrices are symmetric by construction and the diagonal elements are equal to 1. Darker shades correspond to higher co-clustering probabilities.

5.2.0.1 PPM posterior similarity. The PPM matrix (left panel) exhibits a diffuse and granular structure. Although a block pattern is visible along the diagonal, the boundaries between blocks are not sharply defined and many entries display intermediate color intensities. This indicates that numerous pairs of observations have moderate posterior probabilities of belonging to the same cluster. Such a pattern reflects a posterior distribution that is not strongly concentrated around a single partition. Instead, multiple clustering configurations receive non-negligible posterior support. This behavior is consistent with the higher number of clusters and the elevated entropy observed in the MCMC diagnostic plots.

5.2.0.2 PPMx posterior similarity. The PPMx matrix (right panel) exhibits a markedly sharper block-diagonal structure. Within cluster regions appear darker and more homogeneous, while off-diagonal areas are lighter, indicating lower co-clustering probabilities between observations belonging to different groups. This pattern implies stronger posterior agreement in cluster assignments and a higher concentration of posterior mass around a smaller set of partition structures. The clearer separation between blocks reflects reduced ambiguity in the inferred clustering configuration.

5.2.0.3 Comparison and coherence with previous diagnostics. The structural differences between the two matrices are substantial. While the PPM matrix displays widespread intermediate similarity values (indicating posterior fragmentation and variability), the PPMx matrix shows stronger within cluster cohesion and clearer separation across clusters. These results are fully coherent with the MCMC diagnostic plots presented previously. In particular, the higher entropy and greater variability in the number of clusters under the PPM model correspond to the diffuse and less sharply defined block structure observed here. Conversely, the lower entropy and stabilized cluster count under PPMx are reflected in the more concentrated and structured similarity matrix. Overall, the posterior similarity analysis confirms that incorporating covariate information reduces posterior uncertainty over partitions. The posterior variance of the clustering configuration decreases under PPMx, and posterior mass becomes concentrated around fewer, more coherent and stable partition structures.

5.3 Spatial Clustering Comparison

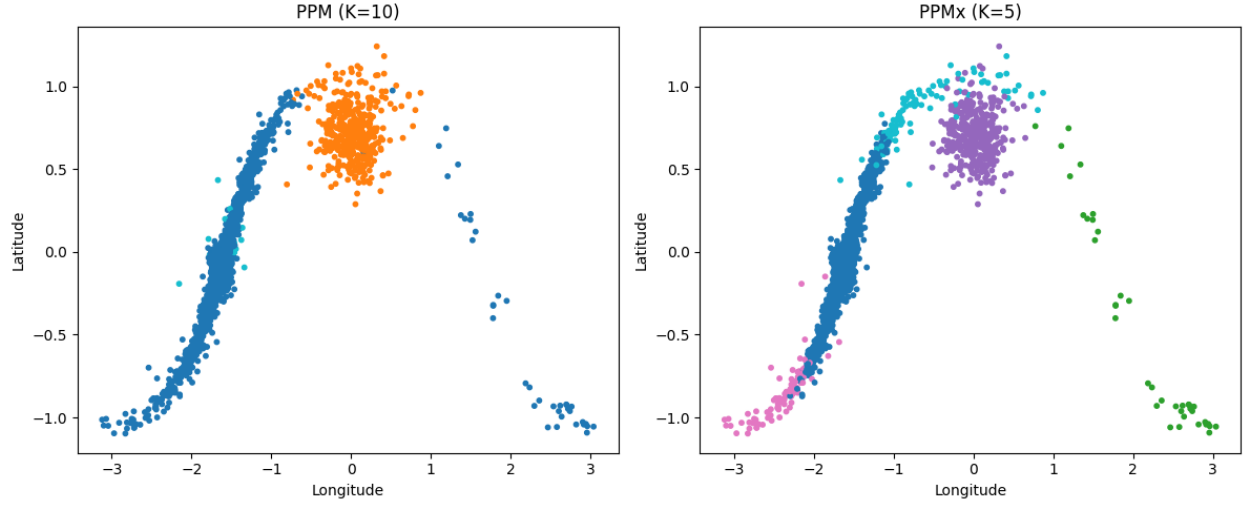


Figure 6: Spatial clustering results in ecliptic coordinates. Left: PPM clustering ($K = 10$). Right: PPMX clustering ($K = 5$). Colors indicate cluster membership under the Binder loss estimate.

5.3.0.1 PPM spatial structure ($K = 10$). Under the PPM model (left panel), clustering is driven exclusively by spatial proximity. The large curved structure on the left-hand side is predominantly assigned to a single cluster, despite visible internal heterogeneity along its length. The compact group in the upper-central region is separated, while smaller peripheral groups are isolated as distinct clusters. Some isolated observations appear less structurally organized.

This behavior reflects the tendency of a purely spatial model to follow local density variations. As a consequence, the partition is relatively fragmented ($K = 10$), consistent with the higher entropy and greater posterior variability observed in the MCMC diagnostics.

5.3.0.2 PPMX spatial structure ($K = 5$). Under the PPMX model (right panel), clustering incorporates both spatial information and covariate similarity. The large curved structure is now subdivided into more coherent subregions, indicating that covariates help detect latent heterogeneity not captured by spatial proximity alone. Peripheral groups are more clearly separated and cluster boundaries appear smoother and more structured. Importantly, the reduction to $K = 5$ does not correspond to a loss of structural detail. Rather, it reflects a more organized partition in which clusters are internally coherent and externally well separated.

5.3.0.3 Comparison and coherence with previous diagnostics. The differences between the two spatial configurations are substantial. While the PPM partition appears more fragmented and locally driven, the PPMX configuration exhibits clearer structural organization and more compact groupings.

These spatial results are fully consistent with the previous diagnostics. The lower entropy, reduced variability in the number of clusters, and sharper posterior similarity matrix observed under PPMX are reflected here in the more stable and interpretable spatial partition. Overall, incorporating covariate information reduces structural fragmentation and decreases posterior uncertainty over partitions. The posterior mass becomes concentrated around fewer, more coherent clustering configurations, leading to improved stability and interpretability of the inferred stellar groups.

5.4 Cluster Structure in Covariate Space (PPMX)

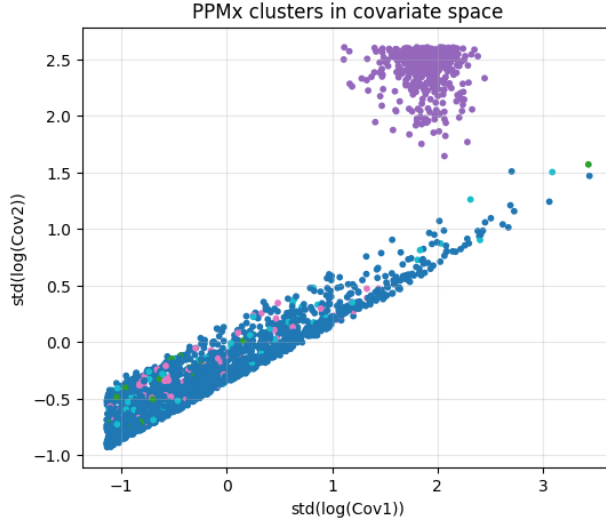


Figure 7: The figure above shows the PPMX clustering projected onto the standardized covariate space ($\text{std}(\log(\text{Cov1})), \text{std}(\log(\text{Cov2}))$). Each point corresponds to an observation, and colors represent the cluster assignment obtained under the Binder loss estimate.

5.4.0.1 Observed structure. A clear structural separation is visible in covariate space. In particular, a compact and well separated group appears in the upper-right region, characterized by systematically higher values of both covariates. The main elongated structure in the lower-left region follows a gradual covariate gradient, within which subregions correspond to distinct clusters.

Cluster boundaries align with changes in covariate magnitude rather than being driven solely by spatial proximity. This confirms that covariate similarity contributes substantively to the clustering mechanism.

6 Prior Sensitivity Analysis

In this section we investigate the sensitivity of the clustering results to different prior specifications using a smaller subset of the complete dataset. Since Bayesian nonparametric models can be influenced by the choice of prior scale, we examine how varying the prior settings affects the posterior number of clusters, partition entropy, and overall structural coherence of the inferred groups. The objective is not to optimize the model, but to assess the robustness of the posterior partition and verify that the main conclusions remain stable under reasonable prior variations. We briefly recall the theoretical role of the main hyperparameters. The parameter α governs the prior tendency to generate new clusters, d affects the distribution of cluster sizes, and η (in PPMX) controls the strength of covariate driven cohesion in the partition.

6.1 Effect of the Concentration Parameter α

The concentration parameter α controls the prior probability of creating new clusters. In both the Dirichlet Process ($d = 0$) and Pitman-Yor process ($d > 0$), larger values of α increase the probability that an observation initiates a new cluster.

As a consequence, increasing α generally leads to:

- A larger expected number of clusters K ,
- Greater partition fragmentation,
- Higher entropy of the clustering configuration.

Conversely, smaller values of α favor fewer and larger clusters, producing more parsimonious partitions.

6.2 Effect of the Covariate Strength Parameter η (PPMX only)

In the PPMX model, the parameter η controls the influence of covariate similarity in the partition prior. When $\eta = 0$, the model reduces to a standard Pitman-Yor mixture based solely on the response variables.

As η increases, the assignment probability increasingly favors clusters whose covariate values are similar. This typically results in:

- Stronger covariate driven cohesion,
- Reduced posterior uncertainty,
- Potential reduction in the number of clusters.

Very large values of η may cause clustering to be dominated by covariate similarity, potentially overriding structure present in the response space.

6.3 Effect of the Discount Parameter d

The discount parameter $d \in [0, 1)$ distinguishes the Dirichlet Process ($d = 0$) from the more general Pitman-Yor process ($d > 0$). While α controls the overall tendency to form new clusters, d affects the distribution of cluster sizes.

Positive values of d tend to favor partitions with:

- More variability in cluster sizes,
- A small number of large clusters,
- Several small clusters (heavy-tailed behavior).

Thus, d influences the structural shape of the partition, rather than only the expected number of clusters.

Empirical Sensitivity Results

In what follows we summarize the empirical evidence from the sensitivity analysis. All figures are reported in the appendix following the common layout described earlier (MCMC diagnostics, PSM, spatial clustering, covariate space).

6.3.0.1 Variation of η (covariate strength). The effect of η is clearly visible in the MCMC diagnostics (Figures A1–A4). For the lowest value ($\eta = 0.35$, A1), the covariate similarity term has limited influence, and the posterior is still largely driven by the response variables. As a result, the sampled number of clusters remains relatively high and shows noticeable variability across iterations (Figure A1), with entropy being less smooth than in the intermediate regimes. For intermediate values ($\eta = 0.5$ and $\eta = 1$, A2–A3), the posterior distribution becomes more concentrated: the number of clusters decreases and stabilizes, and entropy trajectories become smoother (Figures A2–A3). This indicates reduced fragmentation, stronger posterior agreement in allocations, and improved structural coherence. In the extreme case $\eta = 1.5$ (A4), the behavior changes qualitatively. Entropy increases sharply and becomes less stable (Figure A4.1), indicating rising posterior uncertainty. Moreover, both the spatial partition and the covariate-space representation suggest an over-dominance of the covariate similarity term: the covariate-space plot (Figure A4.2) shows cluster boundaries closely following covariate density patterns. This is precisely the regime we want to avoid, since the objective is not to let covariates override the structure induced by the response space. Overall, the empirical evidence suggests that a balanced and stable operating range is approximately $\eta \in [0.35, 1]$, where covariates contribute to coherence without dominating the partition mechanism.

6.3.0.2 Variation of α (concentration). The influence of α is most directly observed through the evolution of the posterior number of clusters (Figures B1–B3). The expected monotonic pattern is confirmed: increasing α leads to larger and more variable values of K . For $\alpha = 0.5$ (B1), the partition is more parsimonious: K is lower and its trajectory is relatively stable (Figure B1). For $\alpha = 1$ (B2), the configuration remains balanced, with moderate cluster counts and stable dynamics (Figure B2). When $\alpha = 2$ (B3), the number of clusters increases visibly and variability becomes more pronounced (Figure B3), indicating stronger fragmentation. This effect is consistent with the PSM structure: as α increases, block patterns become more granular and less sharply separated (Figures B1–B3), reflecting increased posterior mass over more fragmented partitions. Based on the observed diagnostics and spatial structure, α in the range $[0.5, 1.5]$ appears reasonably robust. Values around $\alpha = 2$ tend to induce excessive fragmentation (too many clusters) and are therefore less desirable in this application.

6.3.0.3 Robustness and practical calibration. Taken together, the sensitivity analysis indicates that the main qualitative conclusions of the study remain stable under reasonable prior perturbations. Moderate values of η and α consistently yield coherent partitions, stable MCMC diagnostics, and interpretable spatial/covariate structures (A2–A3; B1–B2). Only extreme regimes lead to clear distortions:

- Large η (A4) can cause covariate dominance, with entropy increasing sharply (A4.1) and clusters aligning too strongly with covariate density patterns (A4.2), producing a substantially different spatial partition.
- Large α increases fragmentation and produces too many clusters (B3), yielding a less parsimonious and less interpretable partition.

7 Conclusions

This report examined the use of Bayesian nonparametric models to identify latent structures in a large astronomical dataset from the LISA mission, addressing the limitations of purely spatial clustering. By incorporating exogenous physical information, specifically gravitational wave frequency and its derivative, within a Product Partition Model with Covariates (PPMX), the proposed approach extends standard partitioning methods in a physically meaningful way. Exploratory analyses highlighted the heterogeneous and heavy-tailed nature of the data, motivating the adoption of the Pitman–Yor process as the partition prior. Compared to the Dirichlet Process, this choice provided greater flexibility in modeling the diverse cluster-size distributions characteristic of stellar populations. A direct comparison between the standard Product Partition Model (PPM) and the PPMX showed that the inclusion of covariates leads to more parsimonious and stable partitions, with a substantial reduction in posterior uncertainty. The resulting clusters exhibit stronger physical coherence, capturing dynamical substructures such as spiral arms that are poorly resolved or over fragmented by purely simple clustering methods. Robustness analyses confirmed that these findings are stable across a reasonable range of prior parameter values. Overall, this work demonstrates that covariate informed Bayesian nonparametric clustering offers a statistically robust and physically interpretable framework for the analysis of large scale astronomical data. Building on these foundations, future research will focus on the construction of clusters of clusters that is a higher level meta clusters that we obtain by grouping the inferred clusters. This second-stage analysis shifts the focus from individual observations to inter-cluster relationships, with the aim of revealing broader astrophysical structures and simplifying the global interpretation of the complex LISA dataset.

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Appendix: Prior Sensitivity Analysis

Across all experiments, the Normal–Inverse–Wishart hyperparameters remain unchanged. The sensitivity analysis is organized into two separate parts, each designed to isolate the effect of a specific prior component.

Appendix A — Variation of the Covariate Strength Parameter η

In Appendix A we fix

$$\alpha = 1, \quad d = 0,$$

and vary the covariate strength parameter η . This section evaluates how increasing the influence of covariate-driven cohesion affects posterior concentration, cluster stability, and structural coherence.

Appendix B — Variation of the Concentration Parameter α

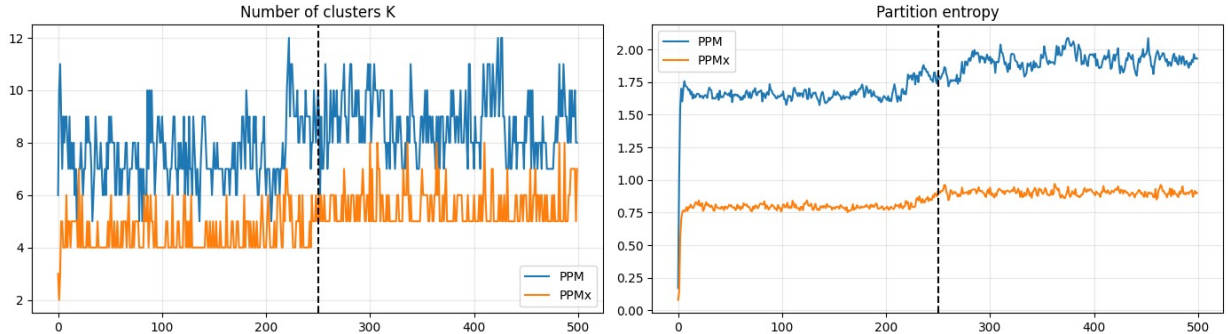
In Appendix B we fix

$$\eta = 0.5, \quad d = 0,$$

and vary the concentration parameter α . This analysis isolates the prior tendency to generate new clusters and evaluates its impact on fragmentation, entropy, and posterior variability.

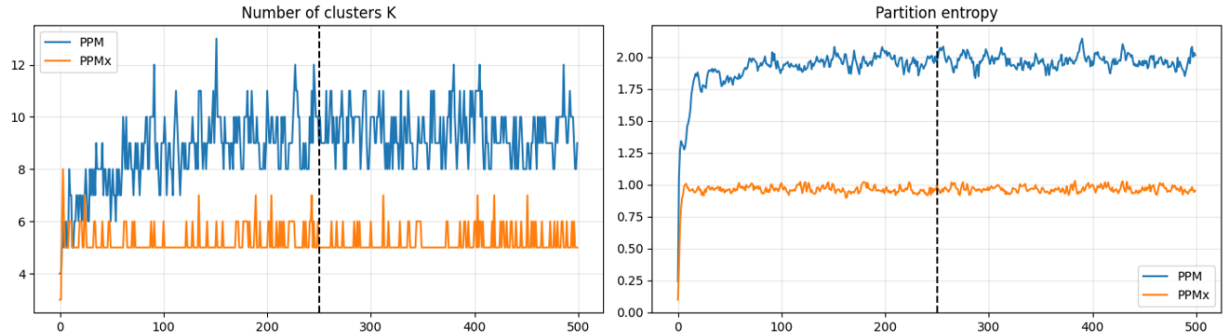
A1 $\eta = 0.35, \alpha = 1$

Figure A1 — MCMC Diagnostics



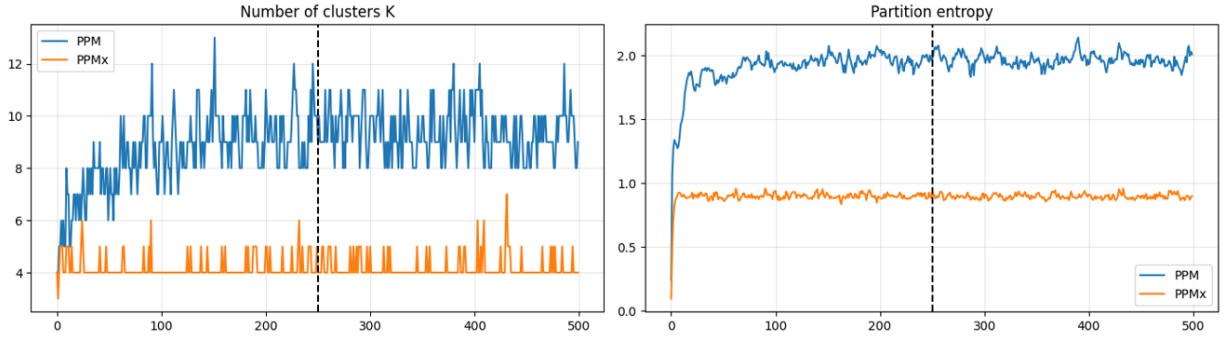
A2 $\eta = 0.5, \alpha = 1$

Figure A2 — MCMC Diagnostics



A3 $\eta = 1, \alpha = 1$

Figure A3 — MCMC Diagnostics



A4 $\eta = 1.5, \alpha = 1$

Figure A4.1 — MCMC Diagnostics

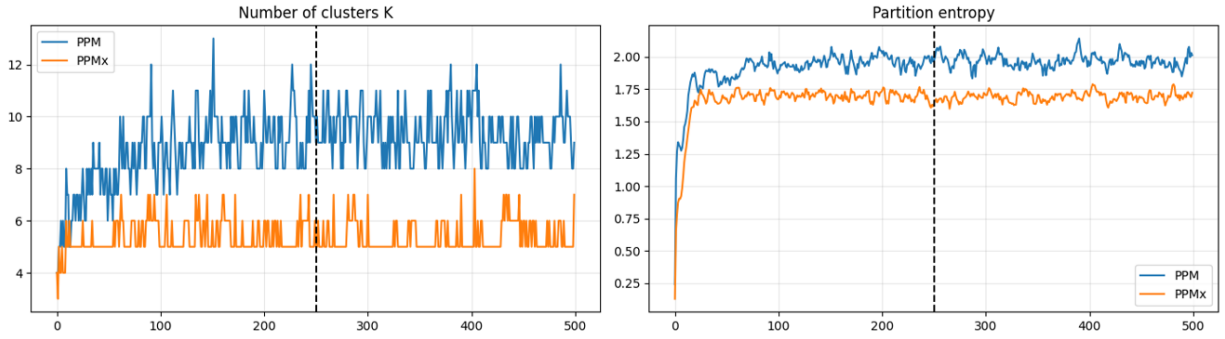
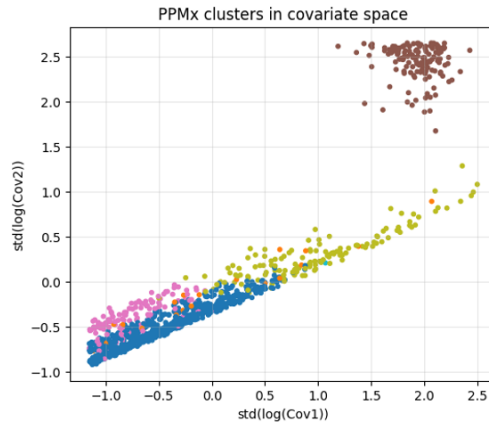
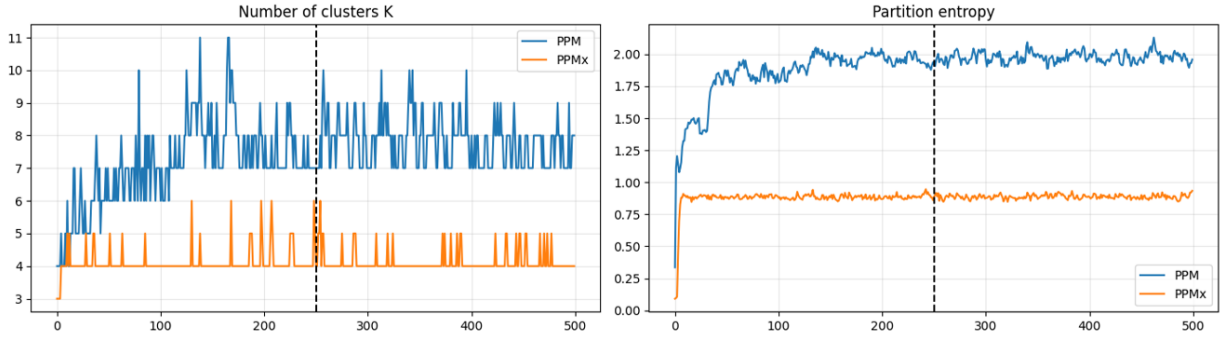


Figure A4.2 — Cluster Structure in Covariate Space



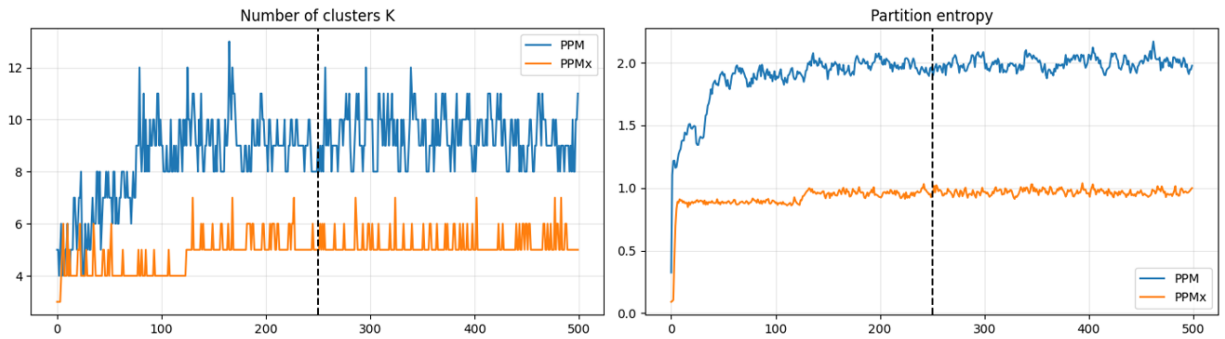
B1 $\alpha = 0.5, \eta = 0.5$

Figure B1 — MCMC Diagnostics



B2 $\alpha = 1, \eta = 0.5$

Figure B2 — MCMC Diagnostics



B3 $\alpha = 2, \eta = 0.5$

Figure B3 — MCMC Diagnostics

